

旋转坐标系中的态矢量为

$$|\psi_{\text{rot}}(t)\rangle = \hat{U}_0^\dagger(t) |\psi(t)\rangle. \quad (8)$$

实验室坐标系中的薛定谔方程为

$$i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = \hat{H}(t) |\psi(t)\rangle. \quad (9)$$

对 (8) 两边求导并代入薛定谔方程，可得旋转坐标系中的薛定谔方程

$$i\hbar \frac{\partial}{\partial t} |\psi_{\text{rot}}\rangle = \hat{H}_{\text{rot}} |\psi_{\text{rot}}\rangle, \quad (10)$$

其中有效哈密顿量为

$$\boxed{\hat{H}_{\text{rot}} = \hat{U}_0^\dagger \hat{H} \hat{U}_0 - i\hbar \hat{U}_0^\dagger \frac{\partial \hat{U}_0}{\partial t}}. \quad (11)$$

推导如下：

$$\begin{aligned} \frac{\partial}{\partial t} |\psi_{\text{rot}}\rangle &= \frac{\partial \hat{U}_0^\dagger}{\partial t} |\psi\rangle + \hat{U}_0^\dagger \frac{\partial |\psi\rangle}{\partial t} \\ &= \frac{\partial \hat{U}_0^\dagger}{\partial t} \hat{U}_0 |\psi_{\text{rot}}\rangle + \frac{1}{i\hbar} \hat{U}_0^\dagger \hat{H} \hat{U}_0 |\psi_{\text{rot}}\rangle. \end{aligned} \quad (12)$$

利用 $\hat{U}_0 \hat{U}_0^\dagger = I$ 求得 $\frac{\partial \hat{U}_0}{\partial t} \hat{U}_0^\dagger + \hat{U}_0 \frac{\partial \hat{U}_0^\dagger}{\partial t} = 0$

$$\frac{\partial \hat{U}_0^\dagger}{\partial t} = -\hat{U}_0^\dagger \frac{\partial \hat{U}_0}{\partial t} \hat{U}_0^\dagger. \quad (13)$$

因此

$$\frac{\partial \hat{U}_0^\dagger}{\partial t} \hat{U}_0 = -\hat{U}_0^\dagger \frac{\partial \hat{U}_0}{\partial t}. \quad (14)$$

代入得

$$\frac{\partial}{\partial t} |\psi_{\text{rot}}\rangle = \left(-\hat{U}_0^\dagger \frac{\partial \hat{U}_0}{\partial t} + \frac{1}{i\hbar} \hat{U}_0^\dagger \hat{H} \hat{U}_0 \right) |\psi_{\text{rot}}\rangle. \quad (15)$$

两边乘以 $i\hbar$ 即得 (11)。

1.2 计算旋转坐标系中的有效哈密顿量

对于 $\hat{U}_0(t) = e^{-i\omega t \hat{S}_z / \hbar}$ ，有

$$\frac{\partial \hat{U}_0}{\partial t} = -\frac{i\omega}{\hbar} \hat{S}_z \hat{U}_0. \quad (16)$$

于是

$$-i\hbar \hat{U}_0^\dagger \frac{\partial \hat{U}_0}{\partial t} = -i\hbar \hat{U}_0^\dagger \left(-\frac{i\omega}{\hbar} \hat{S}_z \hat{U}_0 \right) = -\omega \hat{U}_0^\dagger \hat{S}_z \hat{U}_0 = -\omega \hat{S}_z, \quad (17)$$

因为 $[\hat{S}_z, \hat{U}_0] = 0$ 。